Jorge Enrique Sánchez Encinas

jesanchezencinas@gmail.com | github.com/jsencinas | linkedin.com/in/jsencinas

Education

Tecnológico de Monterrey (ITESM) Expected graduation date: Jun. 2029
B.S. in Computer Science and Technology ***GPA: 3.88***
Relevant Courses: Maths and Data Science (Python), OOP (C++), Data Structures and Algorithms (C++), IoT (SQL, Linux servers), Computer Architecture (Assembly), Software Requirements.

Projects & Experience

HackMTY 2026 Finalist (4th Place) Sep. 2026

Synthia | Real-Time AI Voice Detection Pipeline *Python, Linux, Cloudflare Tunnels, XGBoost*

- Engineered a 3-layer detection architecture combining turn-taking dynamics (51 features), acoustic signal metrics (58 features), and an Out-of-Fold stacked ensemble (XGBoost + Logistic Regression).
- Served the model via a containerized FastAPI endpoint, using Cloudflare Tunnels and a Linux server for continuous uptime and live evaluation.
- Achieved 1.000 ROC-AUC, ~98.8% balanced accuracy, and an average response latency of 1.15 seconds during competition benchmarks.

FIRST Robotics Team Daedalus 7102 Apr. 2023 - Jun. 2025

Lead Programmer & Primary Driver *Java, Git, GitHub, Wpilib*

- Engineered autonomous control systems and led software integration for competition-grade hardware.
- Primary Robot Driver, managing live match operations, strategy execution, and strategic decision-making under high-pressure competition.
- Represented the team at the 2024 FIRST World Championship in Houston, demonstrating advanced robotics programming and effective cross-functional teamwork.
- Key contributor to the Autonomous Award (2024) and Innovation in Control Award (2025) through algorithm development and technical problem solving.

Coordinated based robotic arm May. 2024 - Aug. 2024

Controlled using inverse kinematics in Arduino *Arduino, Python*

- Designed and programmed a 6-degree-of-freedom robotic arm utilizing inverse kinematics to map target (x, y, z) spatial coordinates to joint motor angles.
- Authored modular, object-oriented C++ libraries to optimize code organization, joint interpolation, and hardware abstraction.

Leadership and Volunteering

Techo Jalisco Oct. 2023 - Present

Volunteer lead during housing construction

- Served as primary on-site volunteer lead for the construction of 14 transitional housing units, directing teams of 15 volunteers per build.
- Served as Vice President of ConstruyeTec (2024), supporting strategic planning, project coordination, and campus-wide leadership initiatives.
- Participated in two national fundraising campaigns, driving resource collection and community outreach across Jalisco.

Skills

Programming languages: Java, C/C++, Python, SQL, HTML, CSS, Arduino.
Technologies & Tools: Linux, Zsh, Tmux, Git, Github.
Spoken languages: Spanish (native), English B2.